Beyond Frequent Co-Purchases

A Study of Temporal Stability in Retail Association Rules

Group 11 | Chi Kit WONG (50012338) | Yuxuan OUYANG (50013526) | Zhicheng SUN (50012546)

79.2%
mean monthly retention

46 / 71
rules retained in all four test months

Aug-Nov 2011
held-out evaluation after validation

Question

Do purchasing associations discovered in earlier orders remain reliable in later months?

Data and unit of analysis

541,909 UCI retail line items become **19,773** cleaned baskets; **17,901** UK orders define the primary population. Each invoice is a set of product codes. Missing customer IDs and singleton baskets are retained.

Chronological design

Discover: Dec 2010-May 2011
Select: Jun-Jul 2011
Test: Aug-Nov 2011
Apriori and FP-Growth use the same input and maximum itemset length 3. Parameters are selected before test metrics are computed.

A common retention standard

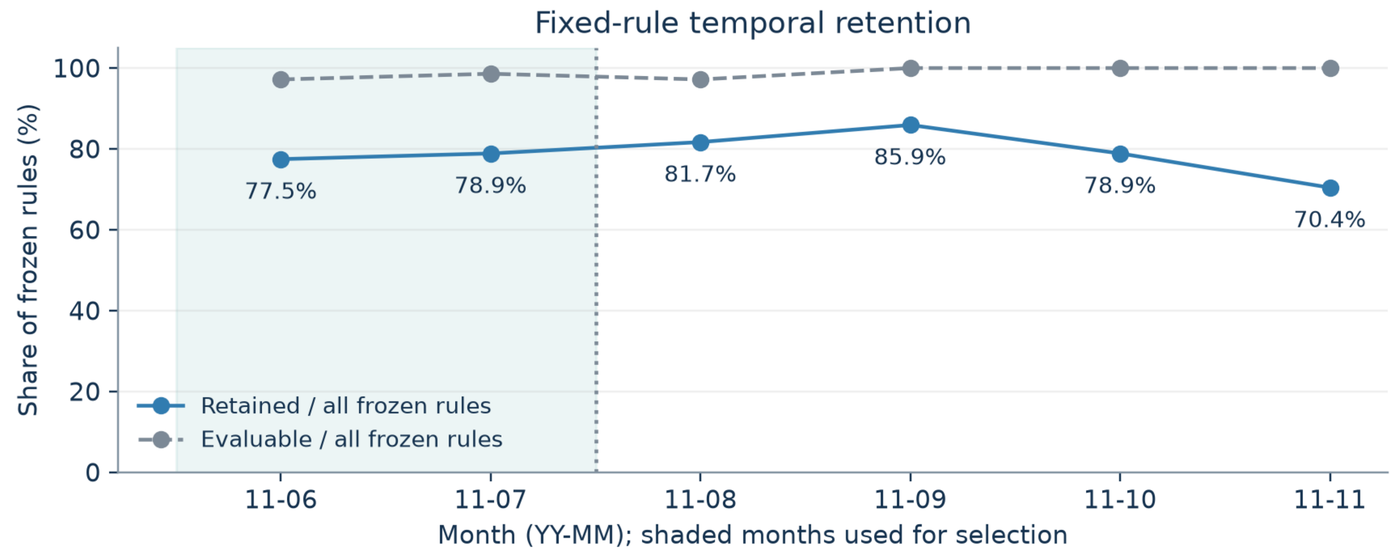
Support at least **0.5%**, confidence at least **50%**, lift at least **1.2**, and at least **20 antecedent orders** per month.
Denominator: all 71 frozen rules. Insufficient evidence is reported separately.

What qualifies the finding

With customer-ID-recorded orders only, retention falls to **64.8%**. Large and unidentified baskets affect the result. A narrower 14-rule setting retains 100%, but is below the predeclared 20-rule floor.

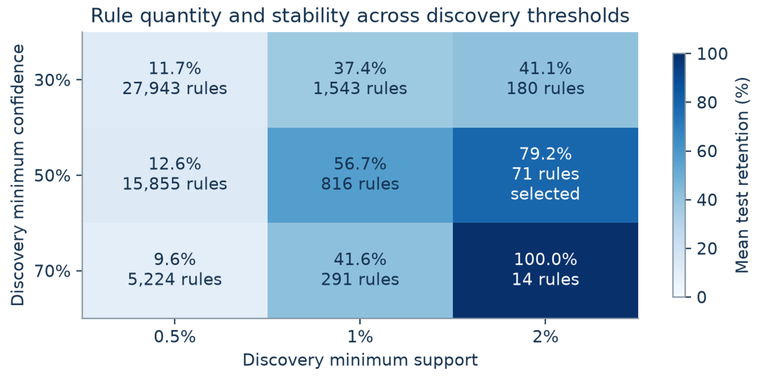
One retailer, four test months, dependent rules and no inventory/promotion history limit generalization. These associations do not establish causation or sales uplift.

Follow the same rules into future months



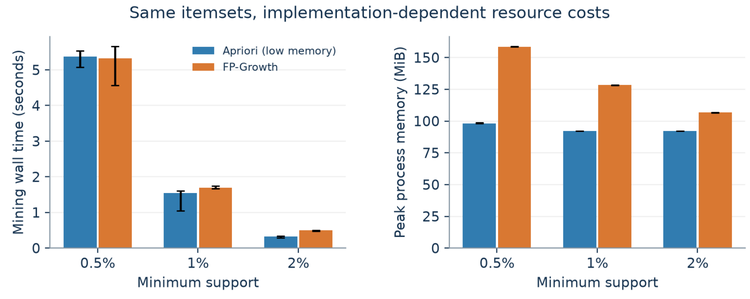
Source: selected UK rules. Common thresholds: support >= 0.5%, confidence >= 50%, lift >= 1.2, antecedent count >= 20.

Quantity and reliability trade off



Source: August-November 2011 UK tests. All cells use the SAME evaluation thresholds.
The selected cell was chosen on June-July validation only; test-grid comparisons are descriptive.

Equivalent mining outputs



Source: 7,365 UK discovery orders; 3 fresh processes per setting. Bars: median; whiskers: min-max.
Mining time excludes loading/encoding; memory includes the Python process and input matrix.

All itemset/count signatures agree. Mining took about **0.3-5.7 seconds** per run, with worker peaks below **160 MiB** on the local CPU. The speed ordering is implementation- and setting-dependent.

Source: Chen (2015), UCI Online Retail, doi:10.24432/C5BW33 (CC BY 4.0). Methods: Agrawal & Srikant (1994); Han et al. (2000); MLxtend.

AIGC: OpenAI Codex assisted implementation, experiments and drafting; team review and revision are required. Full evidence and limitations are in the report.